Mark

Student's names

Signature

Al-Kufa University

Engineering College

Electronic And Communication Department

3rd stage

# LABORATORY MANUAL

# Sequential Logic Circuit Lab

"Laboratory Experiments of Digital Logic Circuits"

## Experiment 5

## Asynchronous Counters

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# 1. Aims:

Analyze and explain the operation of various circuits that is used to design asynchronous counter.

# 2. Theory :

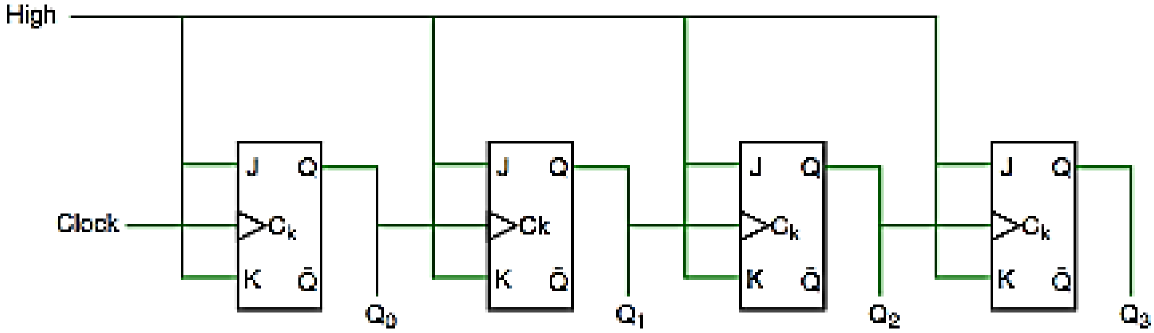
A Counter is a device which stores (and sometimes displays) the number of times a particular event or process has occurred, often in relationship to a clock signal. Counters are used in digital electronics for counting purpose, they can count specific event happening in the circuit.

## 2.1 Classification of counters:

### 2.1.1 According to clock connection:

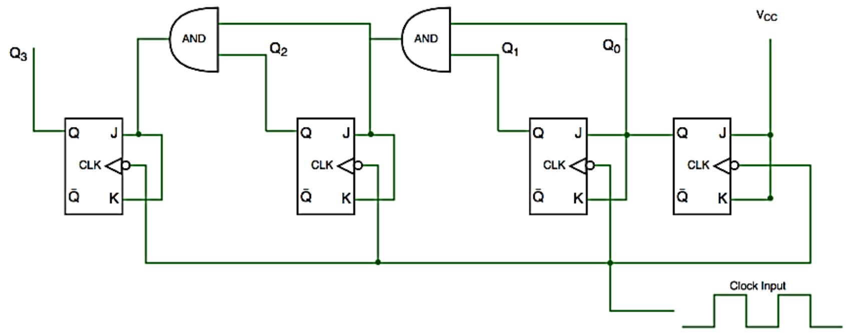
- Asynchronous (Ripple):

In asynchronous counter we don't use universal clock, only first flip flop is driven by main clock and the clock input of rest of the following flip flop is driven by output of previous flip flops. It sometimes called ripple counters because the data appears to "ripple" from the output of one flip-flop to the input of the next. It can easily be made from Toggle or D-type flip-flops.



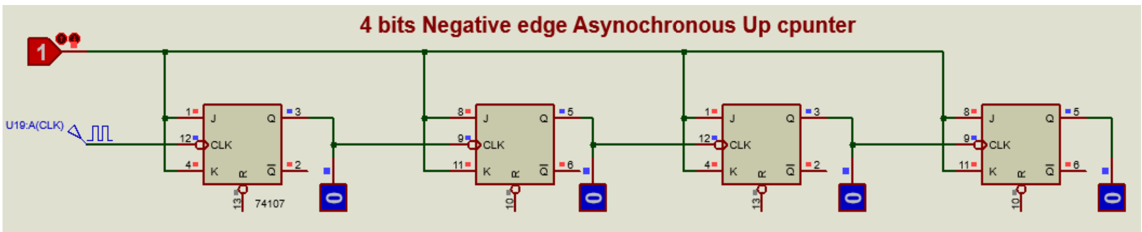

- Synchronous:

Unlike the asynchronous counter, synchronous counter has one global clock which drives each flip flop so output changes in parallel.

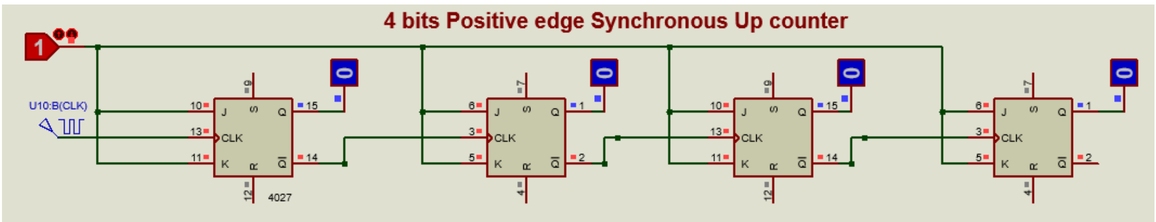



### 2.1.2 According to counter progress:

- Up counter:- can be designed in two ways:
  - ▶ **Negative edge triggered: $Q$ is the clock**




  - ▶ **Positive edge triggered: $Q'$ is the clock**




- Down counter
  - ▶ **Negative edge triggered: $Q'$ is the clock**
  - ▶ **Positive edge triggered: $Q$ is the clock**

To make down counter you can either connect the clk to Q' instead of Q or use the same circuit of up counter and take the output from Q' instead of Q

- Up / Down counter (Bidirectional counter)

It is capable of counting in either the up direction or the down direction through any given count sequence . Generally most bidirectional counter chips can be made to change their count direction either up or down at any point within their counting sequence. This is achieved by using an additional input pin which determines the direction of the count, either Up or Down and the timing diagram gives an example of the counters operation as this Up/Down input changes state.

### 2.1.3 According to number of states:

- ► Fully states: counts from 0 to $2^{number\ of\ states} - 1$
- ► Decade counter (Partial states) (MOD counter)

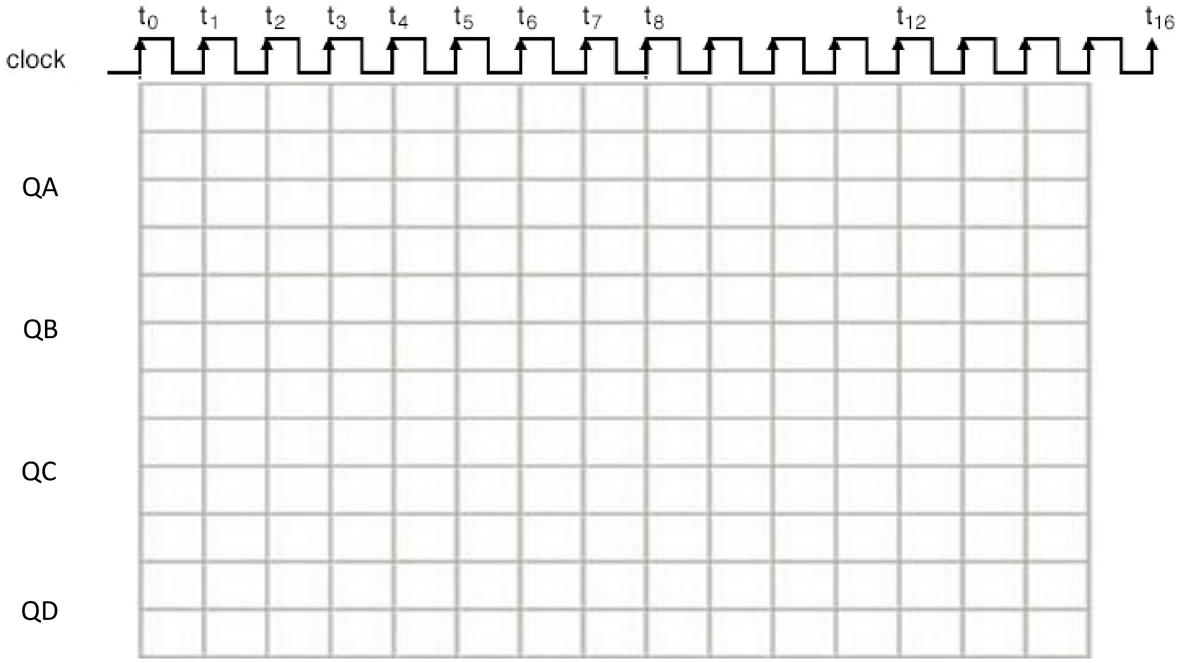
## <u>3. Procedures:</u>

**3.1 Procedure 1:**

1. Recognize the up counter circuits shown in section 2.1.2 and use the same concept to connect 4 bits down counter. Draw the circuit diagram below.

*Note to connect both clr and pre inputs to logic 1*

2. Draw the output results as waveforms on the figure below.




**3.2 Procedure 2 :**

1. Design 4 bits up/down asynchronous counter by filling the table first then connect any combinational circuit you need to clk input. The counter counts up when M = 0 and counts down when M = 1.

| M | Flip flop output | | Output |
|---|---|---|---|
| | $q_0$ | $q_0'$ | clk |
| 0 | 0 | 0 | |
| 0 | 0 | 1 | |
| 0 | 1 | 0 | |
| 0 | 1 | 1 | |
| 1 | 0 | 0 | |
| 1 | 0 | 1 | |
| 1 | 1 | 0 | |
| 1 | 1 | 1 | |

2. Draw the circuit diagram of your design in the figure below.

## 4. Resources

- https://www.geeksforgeeks.org/counters-in-digital-logic/?ref=lbp

74LS90, Decade Counter
74LS92, Divide by 12 Counter
74LS93, 4-bit binary counter
74LS290/74LS196, Decade counter (MOD 10)
74LS293/74LS197, binary counter (MOD 16)
74LS390 Dual decade counter
74LS393 Dual 4-bit counter

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